NLP Course Project Proposals

https://gemini.google.com/app/416d423f018e43dc

User prompt: Predicting stock market from new articles and Price trends Negative Sampling Techniques for Dense Passage Retrieval in a Multilingual Setting News to Numbers: NLP Stock Return Predictions Developed an NLP-driven framework to predict short-term stock returns using financial news headlines. The project integrates pre-trained language models (FinBERT, DeBERTa, OpenAI embeddings) with financial indicators under a bias-free pipeline, achieving 54.4% directional accuracy and a 7.6% Sharpe improvement over traditional baselines SparseFF: Efficient Feed-Forward Sharing in transformers for NLP We propose a Linear Merging Neural Network (LMNN) framework that linearly merges redundant sublayers within transformer blocks to reduce model size and inference cost. Our approach preserves representational fidelity through parameter-space alignment and layer-wise scaling during merging. This enables substantial compression with minimal loss in perplexity and latency, making LMNN efficient for large-scale language models. Semantically Informed Slang Interpretation Can Large Language Models Understand Internet Buzzwords Through User-Generated Content:The paper introduces CHEER, a dataset of Chinese internet buzzwords, and proposes RESS, a method that helps large language models generate more accurate buzzword definitions using user-generated content. Agentic RAG with Multilevel Feedback System This project focuses on developing an Agentic Retrieval-Augmented Generation (Agentic RAG) system that enhances conversational AI through intelligent, multi- step retrieval and reasoning. Instead of relying solely on static context, the agent dynamically identifies missing knowledge, queries both internal and external sources, and refines responses through iterative feedback. A curator LLM structures each interaction into a knowledge base for future retrieval, while the agent module autonomously fetches and integrates external information when internal data is insufficient. This agent-driven retrieval loop ensures higher contextual accuracy, adaptability to topic shifts, and continuous self- improvement in response generation. Frequency-Weighted Watermarking for Large Language Models (LLMs) This project improves the STA-1 watermarking method by giving more weight to rare tokens and less to common ones. It uses token frequency information to create smarter green and red lists, making the watermark stronger and harder to remove while keeping text quality natural. Context Sentiment Detection using XLM Roberta Introduction of Knowledge-Graph in Hybrid Large Language Model-based Framework for Legal Information Retrieval This project enhances the COLIEE 2025 HybridModel for statute law retrieval by integrating a Knowledge Graph (KG) into its multi-stage pipeline. The KG captures structural and semantic relationships among legal articles, enabling candidate expansion and graph-informed re-ranking. These integrations aims to improve both recall and contextual precision, achieving a better F2-score increase over the baseline. The approach bridges symbolic reasoning and LLM- based retrieval, creating a more explainable and legally consistent framework for AI-driven legal information retrieval. Clinical Text Summarization with LLM-Based evaluation Local RAG: Solving document question answering Problems with LLMs Architected efficient RAG pipeline for PDF documents by combining FAISS dense-vector embeddings with Okapi BM25 sparse retrieval and fine-tuned Transformer LLM decoder for precise, context-grounded answer generation, demonstrating a Q&A prototype that reduces manual lookup effort and improves knowledge access. Mood-based Song Recommendation System A mood-based music recommender system that analyzes the user's mood entirely through their text and chat messages. By applying sentiment analysis and natural language processing (NLP) to the user's input, it identifies emotional cues - such as happiness, sadness, stress, or excitement - and recommends music that aligns with or improves their current mood. This creates a personalized, real-time music experience based solely on how the user expresses themselves in conversation. Triage engine— Structured, Multimodal, Uncertainty-Calibrated Clinical Triage for Reliable Decision Support Building a Triage engine multimodal, uncertainty-calibrated clinical triage system that fuses patient text and vital signs to deliver structured, ontology- linked, and auditable triage outputs. It minimizes hallucinations, quantifies confidence, and enables safe, human-in-the-loop decision support for reliable clinical assessments. An NLP-based system that leverages customized Legal-BERT and adversarial analysis to automatically detect and enhance robustness against unfair clauses in online Terms of Service. Phoenix: Explainable Fact-Checking with Stance Intelligence Phoenix AI is a multimodal, multilingual claim verification system that detects fake or misleading information by analyzing both text and images across multiple languages. Using advanced transformer models, it performs stance detection and evidence highlighting, providing transparent, explainable results for🎣 NLP Course Project – Group & Title Submission (DSL504 / DS504) Dear Students, Please fill in your group details in the

shared spreadsheet linked below: 🔗 https://docs.google.com/spreadsheets/d/1KtbvHTn0yJrD7cgB5r9Vi0l13NYtGNsE0Wn45wpnsvE/edit?usp=drivesdk Instructions: • Form groups of up to 5 members • Enter each member's Name and Roll No. in your group's row • Add your Project Title and a short description (3–5 sentences covering the problem, your approach, and novelty) 📌 Note: last year's projects for reference only — you are NOT required to choose from these. Pick your own original topic. ⏰ Deadline: 29/09/2026 For queries, contact your TA. the above are where the references of the prv year could you suggest some really good ideas that we can implement and also like learn from it and that are really good list like all of them

Response: 1. GraphRAG-Reflect: Dynamic Subgraph Pruning and Multi-Hop Self-Correction for Legal QA

- **Spreadsheet Description:** Standard dense retrieval pipelines struggle on complex, multi-hop legal queries where relevant context is fragmented across statutes, precedents, and amendments. We build a hybrid retrieval framework that constructs an article-level Knowledge Graph from statutory corpora and fuses dense semantic representations ($BGE\text{-}M3$) with personalized PageRank subgraph expansion. An iterative, agentic critique module evaluates the relevance of retrieved subgraphs and dynamically prunes noisy edges before passing the grounded context to a decoder model (e.g., $Llama\text{-}3\text{-}8B$). By framing multi-hop legal retrieval as an active graph pruning and reflection problem rather than static text-chunk retrieval, this approach significantly boosts F_2 ranking scores and minimizes cross-statute hallucinations.
- **Datasets:** COLIEE (Statute Law Retrieval / Legal Case QA), CaseLaw, or Indian Kanoon rulings.
- **Core Tech / Models:** NetworkX / PyG, $BGE\text{-}M3$, $Llama\text{-}3\text{-}8B\text{-}Instruct$, LangGraph.
- **Evaluation Metrics:** Recall@K, F_2-score, Faithfulness via Ragas, Hallucination Rate.

A baseline retrieval-augmented system relies on isolated vector lookups over static document chunks. Integrating dynamic subgraph traversal and reflection bridges the structural gap between fragmented legal provisions:

2. SpecDec-Lite: Entropy-Guided Adaptive Draft Verification for Low-Latency LLM Inference

- **Spreadsheet Description:** Autoregressive generation in large language models creates severe memory bandwidth and latency bottlenecks during real-time edge deployment. We design an adaptive speculative decoding framework pairing a compact draft model (e.g., $SmolLM\text{-}135M$ or $Llama\text{-}1B$) with an open target model ($Llama\text{-}3\text{-}8B$). Instead of maintaining a fixed draft lookahead length k, our pipeline monitors target token prediction entropy on the fly to dynamically expand or truncate draft candidate sequences. This entropy-aware verification avoids wasting verification passes on uncertain generation boundaries, achieving an estimated $2.2\times - 2.8\times$ wall-clock speedup on single-GPU hardware with provably zero loss in target distribution perplexity.
- **Datasets:** ShareGPT, MT-Bench, HumanEval / GSM8K.
- **Core Tech / Models:** Hugging Face Transformers, vLLM / custom PyTorch speculative decoding loop, KV-cache reuse.
- **Evaluation Metrics:** Tokens per second (speedup factor τ), Acceptance Rate (α), Output Perplexity (PPL).

3. FaithProbe: Internal Activation Probing and Conformal Calibration for Zero-Shot Hallucination Detection

- **Spreadsheet Description:** Large language models frequently produce plausible yet fabricated assertions in open-domain question answering without signaling internal uncertainty. We extract intermediate residual-stream activations and attention head projections across transformer layers to train lightweight linear probing heads that detect truthfulness representations at the token level. We then integrate these probe confidence scores into a split-conformal prediction framework that bounds risk and outputs mathematically sound confidence sets. Unlike secondary-LLM self-critique methods that double inference latency, our approach operates directly on internal hidden states at inference time, delivering calibrated, guaranteed error bounds ($\alpha \leq 0.05$) on generated responses.
- **Datasets:** TruthfulQA, TriviaQA, HaluEval.
- **Core Tech / Models:** TransformerLens, $Llama\text{-}3\text{-}8B$ / $Mistral\text{-}7B$, Scikit-learn (logistic regression / SVM probes), Non-conformity scoring.
- **Evaluation Metrics:** AUROC, Expected Calibration Error (ECE), Empirical Coverage Rate ($1 - \alpha$).

4. IndicSwitch: Phoneme-Aware Contrastive Alignment for Low-Resource Hinglish Intent & Sentiment Analysis

- **Spreadsheet Description:** Multilingual language models perform poorly on code-mixed social media text (e.g., Hindi written in Roman script) due to irregular phonetic spellings and extreme subword token fragmentation. We construct a dual-stream encoder combining $IndicBERT$ / $XLM\text{-}RoBERTa$ with a character-level phonetic embedding layer trained on synthetic transliteration pairs. Using a multi-task contrastive loss ($InfoNCE$), the network aligns noisy, informal Hinglish utterances and standard formal Hindi/English sentences into a shared, canonical semantic representation space. This architecture circumvents out-of-vocabulary token explosion caused by dialectal spelling variants, yielding significant macro-F_1 improvements on noisy social text without requiring massive manual annotations.
- **Datasets:** DravidianCodeMix, SAIL Hinglish dataset, TweepFake, or IIT Bombay English-Hindi parallel corpus.
- **Core Tech / Models:** $IndicBERT$, $XLM\text{-}RoBERTa$, PyTorch, Soundex/Epitran phonetic featurizers.
- **Evaluation Metrics:** Macro-F_1, Accuracy, Character error rate robustness under synthetic noise injection.

5. AlphaNews: Causal Discourse Graphing and Temporal Cross-Attention for Supply-Chain Stock Return Forecasting

- **Spreadsheet Description:** Existing financial NLP systems treat news headlines as independent sentiment bags of words, failing to capture inter-firm supplier relationships and the multi-day temporal decay of market news. We employ $FinBERT$ and dependency-based open information extraction to automatically construct temporal causal event graphs from financial news feeds and regulatory filings. These structured relational graphs are fed into a Temporal Graph Attention Network ($TGAT$) alongside historical price and volatility series to forecast directional movement over multi-day horizons. By explicitly propagating event shocks along company supply-chain dependencies rather than evaluating sentiment in isolation, our

framework achieves higher directional accuracy and improved backtested Sharpe ratios over classical financial NLP baselines.

- **Datasets:** Financial PhraseBank, Kaggle FNSPID (Financial News and Stock Price Integration Dataset), SEC 10-K relation graphs.
- **Core Tech / Models:** $FinBERT$, SpaCy / OpenIE, PyTorch Geometric (PyG), Yahoo Finance API.
- **Evaluation Metrics:** Directional Accuracy (DA), Cumulative Simulated Return, Sharpe Ratio, Maximum Drawdown (MDD).

6. PromptShield: Latent Trajectory Probing for Robust Indirect Prompt Injection Detection

- **Spreadsheet Description:** Autonomous LLM agents integrated with web search and document retrievers are vulnerable to indirect prompt injection and adversarial goal-hijacking hidden within untrusted external text. We develop a lightweight guardrail detector that monitors trajectory shifts in the LLM's early-layer hidden representations when processing untrusted retrieved context alongside system instructions. By applying causal mediation analysis to isolate attention patterns that divert instruction compliance, the system flags adversarial command sequences before auto-regressive decoding begins. Rather than relying on fragile keyword blocklists or easily bypassed prompt filters, PromptShield detects structural attention diversion inherent to instruction overrides, remaining resilient against unseen adversarial jailbreak suffixes.
- **Datasets:** BIPIA (Benchmark for Indirect Prompt Injection Attacks), AdvGLUE, JailbreakBench.
- **Core Tech / Models:** PyTorch, Hugging Face Transformers, Representation Engineering toolkit ($RepE$).
- **Evaluation Metrics:** Attack Success Rate (ASR), Benign Prompt False Positive Rate (FPR), Latency overhead (ms).

7. DecompSense: AST-Constrained Semantic Reconstruction and Vulnerability Explanation for Stripped Binaries

- **Spreadsheet Description:** Reverse-engineering compiled assembly code into readable, secure source code remains challenging due to stripped symbol tables, compiler optimizations, and lost variable abstractions. We construct a transformer-based decompiler fine-tuned on disassembled binary blocks ($x86\text{-}64$) paired with decompiled C functions from open-source repositories. We enforce grammatical Abstract Syntax Tree (AST) mask constraints during decoding to guarantee syntactic validity, coupled with an auxiliary classification head that scans generated code for common memory safety vulnerabilities. Incorporating formal grammar constraints directly into the token generation step eliminates syntax hallucinations, achieving higher compilation pass rates and producing actionable security audit explanations for binary analysts.
- **Datasets:** AnghaBench, GitHub C vulnerability repos (CWE-119, CWE-120), Ghidra/IDA Pro disassembled corpora.
- **Core Tech / Models:** CodeLlama / DeepSeek-Coder-$1.3B$-$6.7B$, Tree-sitter AST parser, Capstone disassembly engine.
- **Evaluation Metrics:** CodeBLEU, Pass@K (syntax compilation rate), Vulnerability Detection Accuracy (F_1).

8. MedCalib: Uncertainty-Calibrated Multimodal Triage with Evidence-Grounded Counterfactual Explanations

- **Spreadsheet Description:** Automated clinical triage models frequently suffer from overconfidence and lack transparent, verifiable attribution to raw clinical notes. We propose an uncertainty-calibrated triage engine that fuses free-text clinical triage notes (processed via $ClinicalBERT$ / $BioMistral$) with numeric emergency vital signs using cross-modal transformer fusion. We calibrate risk predictions using temperature scaling and conformal prediction while leveraging integrated gradients to generate counterfactual reasoning paths (e.g., highlighting specific vital shifts or symptom phrases that altered the acuity level). This framework provides clinically auditable, uncertainty-bounded decision support that quantifies model doubt and safely triggers human clinician escalation when confidence falls below safety thresholds.
- **Datasets:** MIMIC-IV-ED (Emergency Department), PhysioNet Challenge datasets.
- **Core Tech / Models:** $BioLinkBERT$, PyTorch, Captum (Integrated Gradients), Conformalized Quantile Regression.
- **Evaluation Metrics:** Quadratic Weighted Kappa, Expected Calibration Error (ECE), Brier Score, AUROC under deferral.

Project Selection Matrix

Project Title	Primary NLP Domain	Technical Complexity	Minimum Compute Needed
GraphRAG-Reflect	Retrieval & Knowledge Graphs	Moderate	Single T4 / RTX 3090
SpecDec-Lite	Systems NLP & Inference Optimization	High	Single RTX 3090 / A100
FaithProbe	Interpretability & Hallucination	Moderate	Single T4 / Colab Free Tier
IndicSwitch	Multilingual & Code-Mixing	Moderate	Single T4 / Colab Free Tier
AlphaNews	Financial NLP & Graph Neural Networks	Moderate-High	Single T4 / RTX 3060+
PromptShield	AI Safety & Adversarial Robustness	Moderate	Single T4 / Colab Free Tier
DecompSense	Code Intelligence & Constrained Decoding	High	Single RTX 3090 / A100
MedCalib	Clinical NLP & Uncertainty Calibration	Moderate	Single T4 / RTX 3090

User prompt: list more ideas in really good way

Response: Here are seven high-impact, research-grade NLP project proposals designed for your course submission spreadsheet, targeting current frontiers in reasoning verification, mechanistic steering, parameter-efficient inference, and preference alignment.

1. NeuroSteer: Representation-Level DPO and Activation Steering for Surgical Concept Erasure

- **Spreadsheet Description:** Standard Direct Preference Optimization (DPO) and fine-tuning methods for model unlearning frequently cause catastrophic forgetting across unrelated domains and degrade core reasoning benchmarks. We propose a localized concept erasure framework that identifies target concept subspaces within intermediate residual streams using linear discriminant analysis on hidden activations. We then train low-rank directional steering vectors optimized via representation-constrained DPO to penalize generations that project onto the target subspace while enforcing a strict KL-divergence penalty on benign prompt completions. This enables surgical erasure of copyrighted text, personally identifiable data, or hazardous chemical synthesis knowledge from open models like $Llama\text{-}3\text{-}8B$ with minimal disruption to general language understanding.
- **Datasets:** TOFU (Task of Fictitious Unlearning), WMDP (Weapons of Mass Destruction Proxy), TruthfulQA, MMLU.
- **Core Tech / Models:** $Llama\text{-}3\text{-}8B\text{-}Instruct$, PyTorch, TransformerLens, Hugging Face TRL (DPO trainer).
- **Evaluation Metrics:** Unlearning Efficacy ($1 - \text{ROUGE}$ / Perplexity on forgotten subset), Retained General Knowledge (MMLU accuracy Δ), Representation Cosine Divergence.

Direct Preference Optimization bypasses separate reward model training by directly penalizing target tokens, but unlearning benefits substantially when paired with internal activation steering:

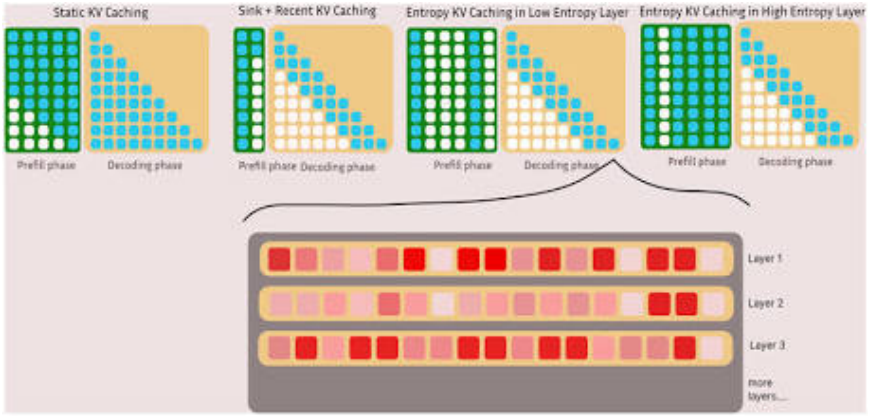
2. AdaCache: Attention-Entropy Budgeting and Dynamic KV-Cache Eviction for Ultra-Long Context RAG

- **Spreadsheet Description:** Long-context Retrieval-Augmented Generation (RAG) is bounded by the linear growth $\mathcal{O}(N)$ of Key-Value (KV) cache memory, causing memory exhaustion during multi-document synthesis on single-GPU hardware. We build a dynamic KV-cache eviction policy that tracks attention-matrix entropy $\mathcal{H}_l$ and token activation frequency across attention heads at every generation step. Tokens exhibiting low persistent attention density are compressed into lightweight centroid approximations or dropped entirely, while initial prompt tokens ("attention sinks") and high-entropy pivotal tokens are preserved at full float precision. This dynamic budgeting reduces runtime KV-cache memory footprints by $60\%–75\%$ during $32k–64k$ token retrieval generation while retaining high retrieval faithfulness.
- **Datasets:** Needle In A Haystack, LongBench, NarrativeQA, MultiFieldQA.
- **Core Tech / Models:** $Mistral\text{-}7B\text{-}Instruct$, FlashAttention-2, PyTorch custom attention masks, vLLM / Hugging Face.
- **Evaluation Metrics:** Memory Footprint (GB VRAM allocated), Retrieval Accuracy@K, Output Perplexity (PPL), Generation Latency (ms/token).

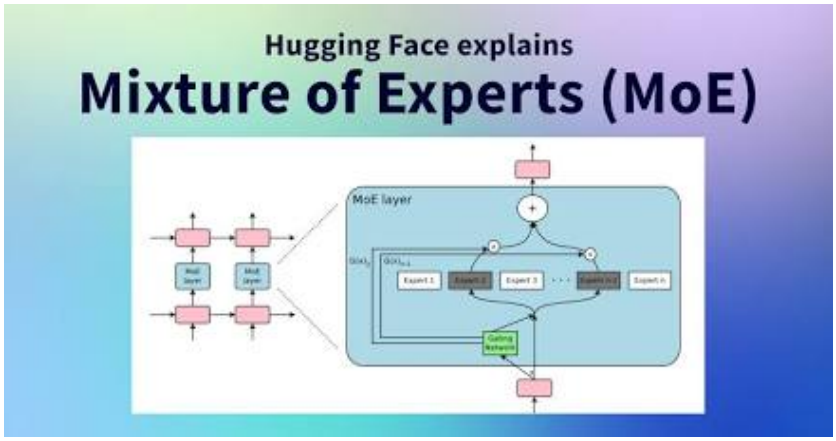
3. RouterGuard: Entropy-Regularized Dynamic Routing and Expert Pruning for Sparse MoE Architectures

- **Spreadsheet Description:** Sparse Mixture-of-Experts (MoE) models achieve high parameter scale at lower inference costs, but real-world routing networks suffer from expert specialization collapse where a small fraction of experts dominate token processing. We introduce an entropy-regularized routing mechanism that injects token-level load balancing directly into top-k gating networks to ensure uniform expert distribution across multilingual and domain-specific tokens. Furthermore, we develop a post-training importance score that identifies redundant, overlapping expert sub-networks and prunes them without structural retraining. Our approach demonstrates that a compact open MoE (e.g., $Qwen$-1.5-MoE or $OLMoE$-1B-7B) can be pruned by up to 30% of active expert weights while maintaining zero degradation on downstream tasks.
- **Datasets:** C4 / Dolma multilingual corpora, GSM8K, ARC (Abstraction and Reasoning Corpus).
- **Core Tech / Models:** $OLMoE$-1B-7B, $Qwen$-MoE, PyTorch custom gating kernels, Hugging Face Accelerate.
- **Evaluation Metrics:** Expert Utilization Entropy (H_{routing}), Task Accuracy across STEM and commonsense benchmarks, Inference Speedup (τ).

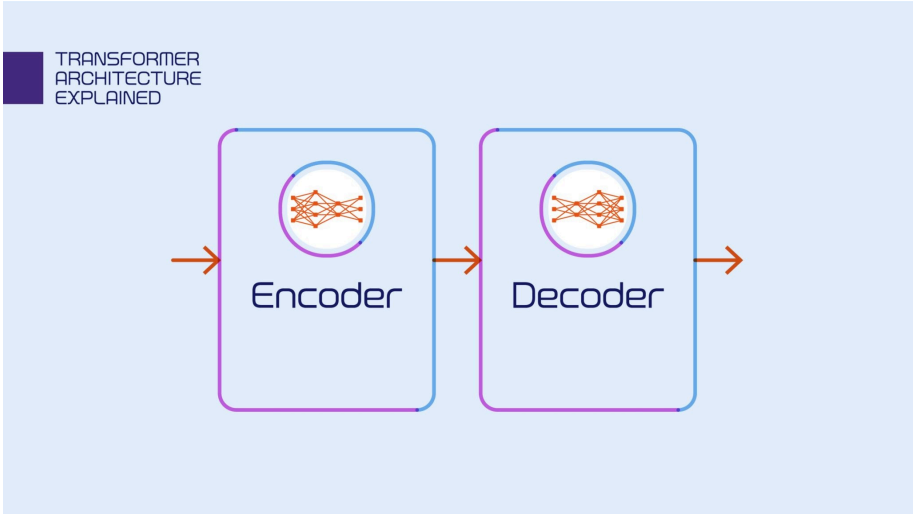
The gallery below highlights three core structural paradigms in modern transformer optimization: managing memory bandwidth via KV-cache pruning, dynamic gating in Mixture-of-Experts layers, and standard encoder-decoder representations.



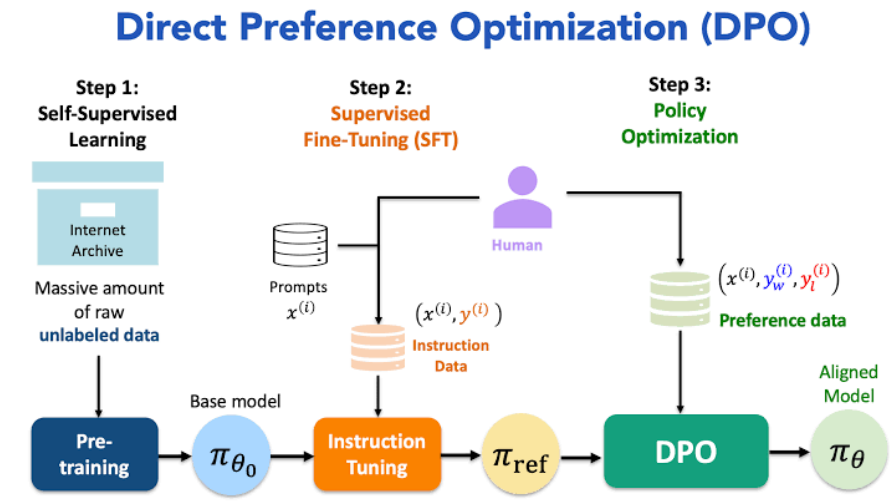
Entropy-Guided KV Caching for Efficient LLM Inference · MDPI



Mixture of Experts Explained · Hugging Face



atdigit / Getty Images



Direct Preference Optimization (DPO) of LLMs: A Paradigm Shift ... · Medium

4. StepVerify: Early-Exit Token-Level Process Reward Probing for Multi-Step Mathematical Reasoning

- **Spreadsheet Description:** Chain-of-Thought (CoT) reasoning models frequently produce subtle logical errors in intermediate steps that cascade into unrecoverable final answers, yet running secondary verifier LLMs over every step introduces unsustainable inference latency. We develop a token-level Process Reward Probe that extracts internal hidden states and self-attention confidence scores directly from the primary model's intermediate layers. By training a lightweight linear classifier on layer-wise residual vectors at newline or step-boundary tokens, the model detects step derailments in real time ($< 5\text{ms}$ overhead). When step confidence falls below a calibrated decision boundary θ, the decoder halts forward rollout and triggers targeted single-step backtracking rather than re-generating entire solutions from scratch.
- **Datasets:** PRM800K, GSM8K, MATH, SVAMP.
- **Core Tech / Models:** $DeepSeek\text{-}Math\text{-}7B$, $Llama\text{-}3\text{-}8B$, PyTorch, TransformerLens / Scikit-learn.
- **Evaluation Metrics:** Step-level Precision/Recall (F_1), End-to-End Solve Rate on MATH, Verification Compute Overhead (FLOPs / Token latency).

5. CultureAlign: Cross-Lingual Contrastive DPO for Low-Resource Regional Pragmatics and Norms

- **Spreadsheet Description:** Foundation language models predominantly align with Western and Anglocentric social heuristics, failing to respect context-dependent honorifics, indirect speech acts, and localized moral norms common in Indian regional dialogues. We curate a culturally contextualized preference dataset covering social scenarios, etiquette disputes, and ambiguous ethical dilemmas in English, Hindi, and Hinglish. We fine-tune a multilingual model (such as $Sarvam\text{-}2B$ or $IndicLLM$) using a cross-lingual contrastive DPO objective that aligns latent social representations across translations without altering factual capability. This framework demonstrates how small, regional open models can avoid cultural erasure while sustaining high task performance across non-English instructions.
- **Datasets:** NormBank, Social Chemistry 101 (extended with synthetic and manually curated Indian cultural scenarios), Flores-200.
- **Core Tech / Models:** $Sarvam\text{-}2B$, $Llama\text{-}3\text{-}8B\text{-}Instruct$, Hugging Face TRL, PyTorch.
- **Evaluation Metrics:** Cultural Alignment Win Rate (LLM-as-a-judge + human annotation), Cross-Lingual Consistency Score, Perplexity on benchmark tasks.

6. FormalSynthesize: SMT-Constrained Multi-Agent Self-Play for Provably Correct Code Synthesis

- **Spreadsheet Description:** While Large Language Models can write syntactically clean code, functional correctness on complex boundary cases and loop invariants remains unreliable. We design a multi-agent self-play architecture where a Generator Agent drafts Python functions alongside formal pre- and post-condition assertions, while an Adversarial Prover Agent generates edge-case counterexamples. An underlying SMT solver (Z3) checks the generated code against formal constraints to mechanically detect invalid assumptions before unit testing. The generator iteratively refines code based on concrete counterexample traces produced by the solver, eliminating logical corner-case bugs without relying on expensive manual unit-test authoring.
- **Datasets:** HumanEval, MBPP (Mostly Basic Python Problems), LeetCode contest problems.
- **Core Tech / Models:** $DeepSeek\text{-}Coder\text{-}6.7B$ / $Qwen\text{-}2.5\text{-}Coder\text{-}7B$, Z3 Theorem Prover, LangGraph.
- **Evaluation Metrics:** $\mathrm{Pass@1}$, $\mathrm{Pass@5}$, Verification Proof Coverage Rate, Average Refinement Iterations per solved problem.

7. FactAnchor: Entity-Grounded Contrastive Decoding for Real-Time Hallucination Suppression

- **Spreadsheet Description:** Hallucination mitigation in open-domain text generation is largely tackled via post-generation critique, which doubles token generation latency and fails on subtle entity attribution errors. We propose a real-time contrastive decoding framework that modifies token sampling distributions during the decoding phase. At each step t, the system computes an entity-grounded probability distribution by contrasting the base model's logits against an intentionally distorted amateur model conditioned on corrupt entity facts extracted from a dynamic Knowledge Graph:

$$\mathcal{L}_{\text{final}} = \log p_{\text{base}}(y_t \mid x, y_{<t}) - \alpha \cdot \log p_{\text{corrupt}}(y_t \mid x, y_{<t})$$

This penalizes tokens that rely heavily on memorized, ungrounded priors, forcing the model to generate only assertions backed by factual evidence without needing auxiliary re-ranking models.

- **Datasets:** WikiBio Hallucination Dataset, HaluEval, FEVER.
- **Core Tech / Models:** $Llama\text{-}3\text{-}8B$, Wikidata / DBpedia SPARQL API, PyTorch custom decoding hook.
- **Evaluation Metrics:** Factuality Score (FactScore), Hallucination Rate, Average Latency per token, BERTScore.

Project Feasibility & Scope Comparison

Project Title	Primary Focus Area	Implementation Complexity	Primary Hardware Requirement
NeuroSteer	Mechanistic Alignment & Unlearning	Moderate	Single $T4$ / Colab Pro (16GB)
AdaCache	Systems NLP & Inference Optimization	High	Single $RTX\ 3090$ / $A100$ ($24\text{GB}+$)
RouterGuard	MoE Architectures & Pruning	High	Single $A100$ (40GB) / Kaggle Multi-GPU
StepVerify	Mathematical Reasoning Verification	Moderate	Single $T4$ / Colab Free Tier
CultureAlign	Multilingual Alignment & Sociolinguistics	Moderate	Single $RTX\ 3060$ / $T4$ (16GB)
FormalSynthesize	Code Intelligence & Formal Methods	Moderate-High	Single $T4$ / API-based local runtime
FactAnchor	Decoding Algorithms & Knowledge Graphs	Moderate	Single $T4$ / Colab Free Tier

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